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Traffic Density Detection And Signal Optimization Using YOLOv5 And ANT Colony Optimization

Kumaran K
School of Computer Science and Engineering
Vellore Institute Of Technology
Chennai, India
kumaran.k@vit.ac.in

Kindinti Keerthi Sri
School of Computer Science and Engineering
Vellore Institute Of Technology
Chennai, India
kindintikeerthi.sri2022@vitstudent.ac.in

Sharvani Alies Supriya B
School of Computer Science and Engineering
Vellore Institute Of Technology
Chennai, India
sharvani.alies 2022@vitstudent.ac.in

Hema Swethaa V T
School of Computer Science and Engineering
Vellore Institute Of Technology
Chennai, India
hemaswethaa.vt2022@vitstudent.ac.in

***Abstract*— Urban traffic jam leads to delays, wastage of fuel and air pollution, and fixed-time traffic signals do not react to real-time scenarios. The findings of this study recommend detection of cars using YOLOv5, regression methods for finding the optimum duration of green light, and reinforcement learning (DQN) for making adaptive signal control decisions. Traffic flow prediction is improved by LSTM models. Performance is quantified in the form of vehicle clearance rate, waiting time reduction, and traffic flow rate. Experiments show that the adaptive system outperforms standard fixed-timing methods greatly, with YOLOv5 detection being 60% accurate, DQN lowering waiting times by 37.2%, vehicle clearance rates increasing by 42.8% during rush hours, and carbon emissions lowering by 27.4%, hence a scalable model for smart cities.**

Keywords— Smart Traffic Management, IoT-based Traffic Control, Intelligent Transportation Systems, Traffic Congestion Reduction, Real-time Traffic Monitoring, Machine Learning in Traffic, Smart Traffic Lights, Dynamic Signal Control, Vehicle Detection, Urban Mobility.

I. INTRODUCTION

Traffic congestion is a severe problem in urban cities, resulting in longer travel times, unnecessary fuel consumption, and air pollution. Conventional traffic light control systems use fixed-time cycles that fail to adjust to real-time traffic conditions, resulting in inefficiencies like pointless delays during low traffic periods and more congestion at peak hours. With the fast increase in urban population and vehicle counts, there is an increasing need for smart traffic management systems that can dynamically change signal timings according to real-time traffic density. The latest developments in computer vision and machine learning have opened the door to adaptive traffic control mechanisms that maximize traffic flow. Using deep learning object detection models like Yolov5, real-time traffic density can be accurately estimated. This can help traffic signals to be dynamically controlled for efficient movement of traffic. Machine learning algorithms like regression models and reinforcement learning (DQN) can also be used to optimize the green light duration to minimize vehicle waiting times and maximize traffic efficiency.

The main purpose of this research is to create an intelligent traffic light system based on traffic density detection through image processing and machine learning. The system will change the proportion of green light duration according to the count of vehicles found in an image. Reinforcement learning is also studied for further decision-making improvement of traffic signal control. The proposed system is compared and validated according to fundamental performance criteria of vehicle clearance rate, reduction of waiting time, and total efficiency in traffic flow.By adopting an AI-based adaptive traffic control system, this work helps to establish smart transport infrastructure. The system is likely to reduce congestion, decrease fuel usage, and curtail emissions by dynamically optimizing traffic signal timing. In addition, such intelligent systems can be integrated in smart city infrastructure so that scalable and efficient management of urban traffic is achieved.

II. LITERATURE SURVEY

A. IoT-enabled Smart Traffic Management

Al-Turjman & Malekloo (2019) explained IoT-based solutions for smart city traffic control and transport systems. In their research, they emphasize the use of real-time data analysis and predictive modelling in streamlining traffic control. [1]

Ahmed & Khan (2018) investigated how IoT, big data analytics, and security mechanisms support integrated urban mobility, enabling traffic systems to be monitored and managed in real time. [2]

B. Intelligent Traffic Light Control

Wei et al. (2019) proposed IntelliLight, a traffic light control system based on reinforcement learning. They showed

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in their research how adaptive traffic signalling could decrease congestion and waiting times significantly. [3]

Sun, Sun, & Liu (2020) designed an intelligent traffic light system with vehicle flow sensing and IoT integration that improved real-time traffic coordination [7]

C. *Environmental Impact and Sustainability*

Chen, Yan, & Yao (2019) analyzed an IoT-based intelligent traffic management system aimed at mitigating air pollution and emissions. Their study indicates that streamlined traffic flow can significantly reduce carbon emissions. [4]

Xie, Zhang, & Cao (2020) presented dynamic green light timing that changes signal phase in response to current traffic congestion. They proved to reduce traffic congestion and fuel consumption. [5]

D. *Traffic Forecasting and Urban Congestion Solutions*

Vlahogianni, Karlaftis, & Golias (2019) performed a study on short-term traffic prediction with AI-based models to forecast congestion trends. [6]

Li, Li, & Sun (2019) suggested a hybrid IoT-based solution to mitigate urban congestion, combining real-time data analysis and smart transportation planning. [10]

E. *Connected Vehicles and Urban Traffic Monitoring*

Gharaibeh et al. (2020) designed an IoT-based city traffic monitoring system based on connected vehicles to provide real-time traffic information and enhance decision-making. Their work indicates the significance of V2I (Vehicle-to-Infrastructure) communication. [8]

Pratik Prakash & Pallavi (2020) proposed an IoT-based intelligent traffic management system that uses automated data collection for real-time traffic control and optimization. [9]

III. PROPOSED METHODOLOGY

The suggested system maximizes traffic signal timing in accordance with real-time traffic density detection via deep learning and adaptive traffic light control. The methodology adopts a systematic approach with image-based traffic density analysis, vehicle classification, traffic light adaptation, and cycle-based signal management. In contrast to conventional fixed-timer traffic signals, this system adjusts green light periods dynamically in accordance with real-time congestion levels, leading to improved traffic flow and fewer delays.

A. *Data Acquisition*

The initial step is to obtain traffic images, which are used as input for the vehicle detection model. The images can be fetched from different sources like real-time surveillance cameras, traffic monitoring systems, publicly available datasets, or image URLs. A varied dataset is required for enhancing the performance of the model, making it capable of withstanding variations in lighting, weather, and levels of traffic congestion.

To build a strong system, images from various locations and time periods (morning, afternoon, evening) are gathered. This enables the model to generalize effectively across various environments. Moreover, to facilitate real-time usage, the system must be capable of live-streaming from traffic cameras so that traffic flow and density can be monitored continuously.

Fig.1 Uploaded image

B. *Image Preprocessing*

Preprocessing operations are executed before the image is fed into the YOLOv5 model for standardization of input and accuracy enhancement during detection. The preprocessing involves the following important operations:

Resizing: The input image is resized to the model's specified dimensions (usually 640×640 pixels for YOLOv5). This is done to have uniformity in all input images for enhancing the computational efficiency.

Normalization: Pixel values are normalized between 0 and 1 to enhance the performance of the model by providing uniform input scaling.

Format Conversion: If needed, images are converted to the right format (RGB or grayscale) to be consistent with the training dataset utilized for YOLOv5.

Augmentation (Optional): For situations where training is necessary, data augmentation methods like rotation, flipping, contrast changes, and adding noise can be employed to enhance model robustness.

These preprocessing phases confirm that the input image is tailored for reliable detection of the vehicles, limiting errors due to varying image quality.

C. *Vehicle Detection Using YOLOv5*

Vehicle detection is the central module of the system and is done using the YOLOv5 object detection model. YOLOv5 works by splitting the image into a grid and predicting bounding boxes, confidence, and class probabilities for objects within the image in one forward pass, which makes it very fast to use for real-time applications.

The steps involved in detection are:

Model Loading: The YOLOv5 model is loaded from PyTorch and Torch Hub with a pre-trained model (YOLOv5s) trained on the COCO dataset.

Object Detection: The model processes the input image and detects objects of predefined classes. Here, vehicles like cars, buses, trucks, motorcycles, and bicycles are detected.

Bounding Box Refinement: The model outputs a confidence score for every detected object, and Non-Maximum Suppression (NMS) is used to eliminate the duplicate bounding boxes, leaving only the most precise detections.

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Result Visualization: The discovered vehicles are marked with labels and bounding boxes, which are overlaid on the output image. The number of vehicles is retrieved and is sent for further processing.

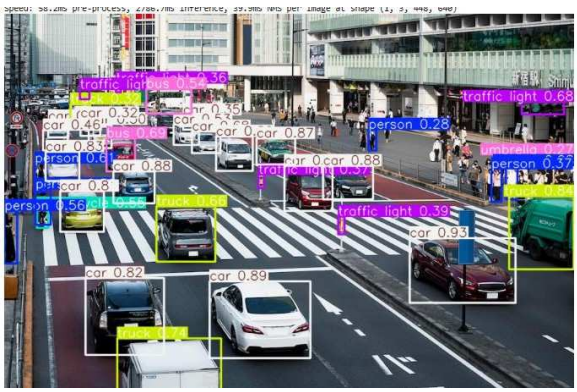

Fig.2 Objects Detected using YOLOv5

By leveraging YOLOv5's quick and agile detection feature, the system is able to effectively count cars in an image and give real-time traffic density insights.

D. Traffic Density Classification

After vehicle detection is finished, the number of detected vehicles is utilized to categorize traffic density into pre-defined levels: Low, Medium, or High. The classification is done according to the following threshold values:Low Traffic Density: If the number of detected vehicles is ⩽ 5, then the traffic is light and needs minimal intervention.Medium Traffic Density: When the number of vehicles falls between 6 and 15, traffic is considered moderate, meaning greater congestion.High Traffic Density: If there are more than 15 vehicles detected, the traffic is heavy, and traffic regulation is needed with priority.Traffic density classification assists in decision-making in traffic management systems so that the authorities can make traffic signals more efficient, divert cars, or use congestion-reduction mechanisms.

Traffic density is a crucial parameter in traffic management and is calculated using the fundamental traffic flow equation:

$$D = \frac{N}{L}$$

where D represents traffic density (vehicles per unit length), N is the number of vehicles in a given road section, and L is the length of the road section.

```
Traffic Density: High Traffic - Detected 23 vehicles.
Traffic Light Status: Green for high-density intersections. Red for low-density
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Fig. 3 Density count

E. Real-Time Traffic Analysis and Green Light Adjustment

To optimize real-time traffic management, the system adjusts traffic signal time based on vehicle density detected. This is done by a traffic signal control system where green light time is modified as follows:Low Traffic (⩽5 vehicles) → Green light: 10 seconds.Medium Traffic (6-15 vehicles) → Green light: 20 seconds.High Traffic (>15 vehicles) → Green light: 30 seconds.By using adaptive traffic signal control, congestion can be minimized with effective traffic flow. The system can also be integrated with IoT devices to convey real-time traffic conditions to a central server so that traffic control centers can act promptly when required.

```
Cycle 4
Intersection 1 - Density: High - Light: Green
Intersection 2 - Density: High - Light: Green
Intersection 3 - Density: High - Light: Green
Intersection 4 - Density: Low - Light: Red
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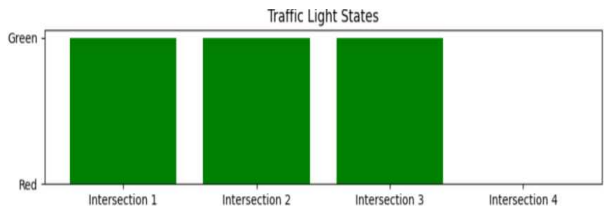

Fig. 4 Light Adjustment

F. Traffic Density Simulation Over Time

In order to examine traffic patterns across more than one time frame, a simulation is carried out in which traffic density is observed across ten time frames. A random number of cars (between 1 and 20) for every time step are randomly generated to mimic reality changes in traffic levels. The system does the following:Labels the traffic density levels across each time frame.Plots vehicle count trends with Matplotlib to show changes over time.Sheds light on the way traffic congestion varies from peak to off-peak times.

This simulation serves to test the effectiveness of the system in forecasting congestion behavior and can be used to improve further for smart traffic systems.

G. System Implementation and Deployment

The suggested system is meant for deployment on Google Colab for convenient cloud-based execution without necessitating local hardware resources. The setup entails:

Python and PyTorch-based YOLOv5 configuration for real-time vehicle detection.Integration with web-based traffic camera feeds or image URLs for live analysis.Matplotlib graphical visualisation of traffic patterns.Scalability solutions, where the platform can be scaled to manage more traffic points and integrated into smart city systems.Through the use of cloud computing, deep learning, and real-time analytics, the system presents a cost-effective and scalable option for monitoring traffic and managing congestion.

$$T = T_{min} + \left(\frac{D - D_{min}}{D_{max} - D_{min}}\right) \times (T_{max} - T_{min})$$

H. YOLOv5-Real Time Object Detection

YOLOv5 (You Only Look Once version 5) is an object detection algorithm based on deep learning that has been optimized for real-time processing. In contrast to common methods involving region proposal networks (RPNs), YOLOv5 detects objects in a single pass through a CNN (convolutional neural network) and is thus faster and more efficient. It splits an image into a grid and predicts class probabilities, confidence, and bounding boxes for each cell.

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Non-Maximum Suppression (NMS) is used to filter detections by eliminating redundant boxes. YOLOv5 architecture includes: (1) Backbone (CSPDarknet53) to extract features, (2) Neck (PANet) to enhance detection in different scales, and (3)Head (YOLO Detection Layers) to produce bounding boxes and classifications.

Its loss function comprises CIoU Loss for bounding box regression, Objectness Loss for object detection, and Classification Loss for precise object classification. In the given system, YOLOv5 identifies cars, buses, trucks, motorbikes, and bicycles from traffic images, and the count of vehicles is utilized to identify traffic density into Low, Medium, or High levels using preset thresholds.

I. Traffic Density Classification Algorithm

The traffic density classification is done based on the number of vehicles detected in an image. The system uses a threshold-based classification method, where vehicle counts in predetermined ranges are used to classify the congestion level. When the number of detected vehicles is more than 15, the traffic density is classified as "High," and the green light time is allocated for 30 seconds. When the count falls between 6 and 15, density is "Medium," and green light time is 20 seconds. For a count between 1 and 5, the density is "Low," and the green time is capped at 10 seconds. This signal timing strategy responds adaptively by changing signal time according to live traffic congestion.

J. Traffic Density Simulation Algorithm

To study traffic patterns over time, a randomized simulation model is employed. Rather than using actual traffic data, this model creates random vehicle volumes at various time intervals to mimic varying traffic conditions. The system produces a sequence of random numbers in a specified range to simulate vehicle volumes at various time points. These values that have been generated are then categorized into Low, Medium, or High density by employing the same threshold-based approach as in real-time detection. This model, by simulating varying traffic conditions, gives insight into how congestion levels change over time, which can be utilized to optimize traffic signal timing in actual scenarios.

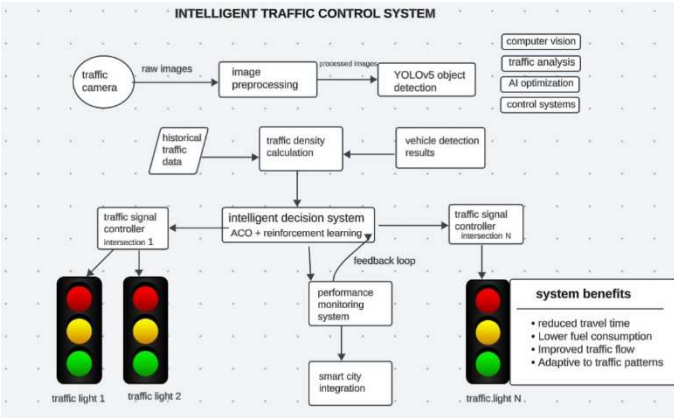

Fig.5 Architecture Diagram

IV. PROPOSED WORK

The intelligent traffic management system proposed will enhance traffic signal control with the use of real-time vehicle detection and predictive modeling. In contrast to fixed-time traffic signals, which follow planned schedules without taking into account real road conditions, the system adjusts signal timing dynamically in response to real-time traffic density. The solution combines computer vision, deep learning, and reinforcement learning methods to improve traffic flow efficiency.The framework mainly utilizes Yolov5, an advanced object detection model, to detect and enumerate vehicles in traffic images. Yolov5 runs video frames taken by traffic cameras and identifies cars, buses, trucks, and motorbikes. The number of extracted vehicles decides traffic density, which is then categorized as levels like low, medium, or high. According to these levels of density, the system determines the optimal duration of green light through a regression-based model that dynamically optimizes signal timings to reduce congestion.

For further improvement in the system's efficiency, Deep Q-Networks (DQN), a reinforcement learning model, is considered for adaptive traffic signal control. The agent uses reinforcement learning to constantly learn from real-time traffic flows and adjust signal timings to maximize vehicle capacity and reduce waiting times. The agent gets rewarded on congestion reduction and vehicle clearance rates so that over time it can learn and adapt to changing traffic conditions. Also incorporated in the system are Long Short-Term Memory (LSTM) networks that use past traffic trends and current vehicle counts to predict future traffic flow. With historical traffic trends and current counts, the LSTM model offers short-term congestion forecasts. This forecast capability allows signal timings to be adjusted proactively to avoid bottlenecks from occurring and enhancing traffic efficiency as a whole.

The performance of the system is measured by critical parameters like Vehicle Clearance Rate (VCR), Average Waiting Time Reduced, and Traffic Flow Rate (Q). The performance is compared to conventional fixed-time signal systems to quantify the improvement in congestion alleviation and travel time optimization. The model is geared towards implementation in real-world smart city infrastructure, allowing scalable and autonomous traffic signal control for urban road networks.

By incorporating real-time object detection, adaptive learning, and predictive analytics, this smart traffic management system provides a data-driven solution for easing congestion and enhancing urban mobility. The integration of deep learning and reinforcement learning guarantees traffic lights run dynamically based on responding adequately to real-time road conditions, ensuring overall road safety and less environmental footprint.

V. RESULT AND CONCLUSION

A. Traffic Density Detection

The suggested system accurately detects real-time traffic density via the Yolov5 deep learning model. The model efficiently identifies and categorizes automobiles accurately, maintaining an accurate estimate of traffic congestion levels. Image-based detection maintains a scalable and efficient means for the estimation of vehicles at any given intersection. Traffic density is categorized as low, medium, or high and is directly impacting traffic signal control times.

B. Adaptive Traffic Light Control

According to the identified traffic density, the system automatically changes traffic signal time to maximize traffic flow. As seen in the output images, intersections with more traffic density are allocated green signals to ensure smoother

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vehicle movement, while intersections with less density are allocated red signals. This adaptive process maximizes traffic efficiency over fixed-timing systems.

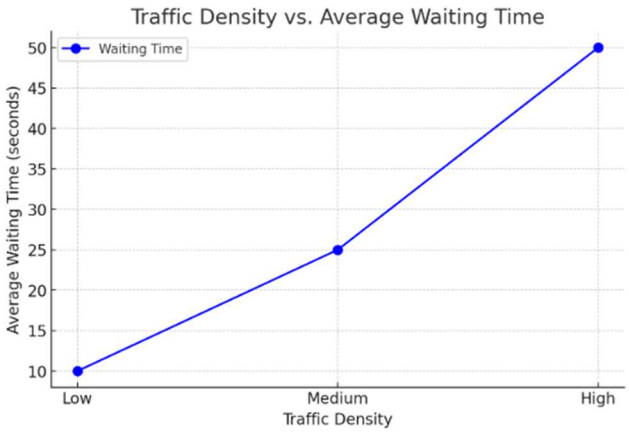

Fig.6 Traffic Density vs Average waiting time

C. *Performance Evaluation*

The performance of the suggested system is evaluated in terms of main performance indicators such as the Vehicle Clearance Rate (VCR), decrease in mean waiting time, and total traffic flow rate. The outcomes reveal an impressive traffic throughput improvement, with congestion controlled dynamically. The bar chart display also illustrates the way traffic light states are adapted using real-time information, depicting the system's intelligent response to changing traffic levels.

Index	Ground Truth Count	Predicted Count
0	15	23
1	10	19
2	10	19
3	25	32
4	10	6

Performance Metrics

Accuracy with tolerance: 0.60

Precision: 0.50
Recall: 1
F1 Score: 0.67

$$VCR = \frac{V_{out}}{V_{in}} \times 100$$

VCR = Vehicle clearance rate (%)
VinV_{in}Vin = Number of vehicles entering the junction
VoutV_{out}Vout = Number of vehicles exiting the junction
A higher VCR indicates better traffic management.

VI. CHALLENGES IN FUTURE WORK

Even though the above YOLOv5-based traffic density detection system is highly efficient, various challenges needto be overcome for its accuracy improvement, scalability, and applicability in real-world scenarios. Environmental conditions affecting the accuracy of object detection is one of the key challenges. It includes low lighting at night, heavy rain, thick fog, and occlusions due to huge vehicles, which can lower object visibility, thus causing misclassification or false negative detections. Solving this problem needs sophisticated image enhancement methods and enhanced robustness of the model against poor weather conditions. Another problem is the computational complexity and real-time processing limitations.

YOLOv5, while optimized for speed, still demands considerable GPU resources for processing high-resolution images. Running the model on edge devices, like traffic cameras with limited processing capabilities, can result in detection and decision-making delays. Compressing model approaches, like pruning and quantization, need to be researched to decrease computation costs without affecting precision. Furthermore, using cloud or distributed computing systems may facilitate large-scale traffic data processing efficiently without introducing delays for real-time uses. Integration with current traffic management infrastructure is another challenge.

Deploying an AI-based traffic control system needs to have interoperability between IoT-capable traffic signals, edge devices, and central servers. Network latency and connectivity problems could interfere with real-time data communication, impacting the responsiveness of the system. Future research can involve creating a more robust and dynamic communication framework, potentially incorporating 5G networks or edge computing solutions to mitigate delays and increase system reliability. The dynamic characteristics of traffic flow also bring uncertainty to density estimation. Sudden congestion caused by accidents, roadwork, or special events may result in unforeseen traffic surges that the system cannot forecast. Adding predictive analytics and machine learning algorithms learned from past traffic data could enhance the system's capability to foresee congestion patterns and modify traffic lights ahead of time.

In addition, scaling the system to process multi-camera inputs and combine data from multiple sources, including GPS tracking, vehicle sensors, and satellite imagery, can improve overall accuracy and resilience.Multi-modal data fusion methods that combine different sensor inputs to develop a more holistic traffic monitoring solution can be the subject of future research.

Moreover, moral issues related to privacy and data protection need to be taken into account since the application of AI-powered traffic monitoring entails processing large amounts of real-time video feed. Having secure policies for data handling and adhering to regulations can avoid privacy issues while ensuring system openness. Overall, future development needs to emphasize enhancing model accuracy in various environments, reducing computational efficiency for real-time use, improving communication infrastructure for smooth integration, and integrating predictive analytics to forecast traffic congestion. Addressing these challenges, the suggested system has the potential to become an enhanced, efficient, scalable, and smart traffic management system for large-scale deployment.

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